International Journal of Communication Networks and Information Security
2024, 16(5)
ISSN: 2073-607X,2076-0930
https://ijcnis.org /
Research Article

Comparative Study of Privacy-Enhancing Approaches in AI Training

Arpit Gadhiraju
Research Scholar, Gujarat Technology University, Vadodara, India
arpit_gadhiraju@gtu.edu.in
Manjunath Panchal
Professor, Gujarat Technology University, School of Engineering, Vadodara, Gujarat, India
sdpanchal@gtu.edu.in

ARTICLE INFO
Received: 05 Oct 2024
Accepted: 06 Nov 2024

ABSTRACT
The accelerating demand for data to support artificial intelligence (AI) model development has made the protection of personal information increasingly complex. Traditional data sharing methods often risk exposure of sensitive details, raising privacy concerns. This research investigates privacy-preserving strategies for machine learning, evaluating approaches such as k-anonymity, differential privacy, noise injection, sampling, and feature selection. Sensitivity analysis reveals that while enhanced privacy safeguards can impact prediction accuracy, careful calibration of model parameters helps minimize these drawbacks. Essential factors considered include computational load, network transmission costs, scalability, and the robustness of privacy guarantees. The paper stresses the importance of finding an optimal balance between protecting data and maintaining model efficacy, advocating for the advancement of algorithms that limit data loss without diminishing performance. The objective is to refine privacy-focused machine learning to achieve equilibrium between protection, utility, and scalability. The core hypothesis posits that the proposed security framework reduces the potential for data leaks and unauthorized disclosures—especially in healthcare, where patient information privacy is paramount. This methodology aspires to support ethical AI progress by harmonizing thorough data exploration with stringent privacy controls.

1.Introduction

The healthcare sector is progressively moving towards digitalization driven by the adoption of advanced technologies and online solutions that enhance user convenience. While these advancements present significant opportunities, they also expose medical organizations to a variety of threats, both digital and nondigital. Such threats have the potential to compromise the security of medical procedures and endanger patient safety [1]. Technical cybersecurity countermeasures now safeguard the confidentiality, integrity, and availability of data and information systems, especially within the healthcare industry. The sector continues to be besieged by various security threats, including insider threats, malware, and DDoS attacks [5]. Unlike other industries, healthcare faces unique challenges. Protected health information (PHI) is particularly valuable, making it a prime target for cybercriminals.

The evolution of data security concerning medical data in India has been significant. Initially, paperbased medical records were common, with security measures relying on physical safeguards like locked cabinets. However, the digitization of healthcare systems in the late 20th century marked a major shift. The adoption of electronic health records (EHRs) [3] introduced both novel opportunities and challenges for data security. While EHRs enhanced efficiency and accessibility, they also heightened concerns about the vulnerability of sensitive patient information.

India, similar to many other countries, has encountered numerous challenges related to the security of healthcare data over the years. Documented breaches in India's healthcare sector have exposed weaknesses in the systems managing sensitive patient information. These breaches often stem from a mix of inadequate cybersecurity measures, outdated technology, and sometimes, malicious activities. Such security breaches have led to increased efforts to improve cybersecurity practices within the healthcare sector. However, they also highlight the persistent challenges in maintaining the security of medical data. Ensuring the protection of confidential patient data in the digital age requires ongoing and coordinated efforts from healthcare organizations, policymakers, and technology experts.

1.1.Cyber Attacks on Healthcare Sector

Cyberattacks on the healthcare sector are an ongoing issue; health data comprise highly detailed personal profiles, rendering them prime targets for fraud, identity theft, and credit card scams.A typical patient's medical record at a healthcare organization includes comprehensive data like the individual's full name, address, date of birth, phone number, email, Social Security (or equivalent) number, emergency contact details, health insurance information, and occasionally even credit card and bank account details. Cybercriminals are indifferent to the size, location, or specific nature of the healthcare provider; their primary objective is to obtain these valuable healthcare records.

Ransomware attacks and data mining efforts have escalated significantly in recent years. Hospitals are often prone to comply with ransom demands because of the critical need to promptly restore medical systems. The healthcare sector's cybersecurity measures lag behind those of other industries, impeded by legacy systems, insufficient digital literacy among staff, and a lack of comprehensive regulations and enforcement mechanisms [5].

A major challenge has been the rapid shift to remote healthcare delivery modalities during the COVID19 pandemic [4,14]. Although advanced technology has enabled healthcare professionals to meet increased demand and deliver routine patient care more safely, it has also prompted organizations to relax security standards to expand telehealth services. This transition necessitated the rapid deployment of new technological platforms, some of which introduced significant security vulnerabilities [10,17].

1.2. Healthcare Data Sharing

Healthcare data sharing takes on various forms and purposes. At its core, it involves patients disclosing their health information with healthcare providers for primary use, essential for delivering medical services. Beyond primary use, researchers, innovators, regulators, and public health experts may seek access to this data for secondary purposes. Another perspective on healthcare data sharing involves distinguishing between individual and aggregate data. Sharing individual data is comparatively straightforward, though challenges such as ensuring access, consent, and interoperability remain significant concerns. In contrast, secondary use often requires aggregating data, which raises critical issues around access and usage, where concerns over data privacy intersect with the requirement for data access and utilization [3].

Despite the widespread adoption of E-health records, health data remains fragmented and isolated, posing significant challenges in accessing data for both primary healthcare needs and secondary uses. Regulatory frameworks have not effectively facilitated data sharing, exacerbating this issue [6]. The debate surrounding the balance between using patient data and safeguarding privacy rights has gained prominence, particularly following the implementation of the EU General Data Protection Regulation (GDPR) [7] in 2018 and the California Consumer Privacy Act (CCPA) in 2020. These regulations have sparked international discussions on the ethical and legal implications of sharing large-scale healthcare data. While earlier laws such as the Health Insurance Portability and Accountability Act (HIPAA) [8] Privacy Rule in the USA and the Personal Information Protection and Electronic Documents Act

(PIPEDA) [9] in Canada provided patients with certain data rights, the GDPR and CCPA have set new standards by enhancing these protections.

The accelerated adoption of telecommuting and telehealth services has amplified concerns regarding data security and compliance with regulations such as GDPR and HIPAA. To effectively detect and mitigate potential privacy breaches, healthcare organizations are required to implement strong access control mechanisms, ensure the safe storage and transfer of patient data, and conduct regular audits and risk assessments. Additionally, there is a need for continuous monitoring and updating of cybersecurity protocols to address emerging threats and maintain regulatory compliance

1.3.Privacy policy in India

India commenced its healthcare digitization efforts through the launch of the National Digital Health Mission, now known as Ayushman Bharat Digital Mission (ABDM) [11]. This initiative aims to establish a comprehensive digital infrastructure to support integrated health services nationwide. The mission is structured around four core components: the Ayushman Bharat Health Account (ABHA), Healthcare Professionals Registry, Health Facility Registry, and Health Information Exchange and Consent Manager (HIE-CM). These elements are designed to identify healthcare providers, professionals, and patients, enabling secure exchange of health data with explicit consent from patients.

A significant challenge currently facing this framework is the absence of dedicated data protection legislation. To address this issue, India has proposed the Data Empowerment and Protection Architecture (DEPA) [12] in draft form. DEPA aims to regulate access to health data by public and private entities through the introduction of 'consent managers'. These managers act as intermediaries between individuals and datarequesting agencies, facilitating data sharing based on individual consent without direct access to the data itself. DEPA draws inspiration from the financial sector's practices in managing data access for rural individuals and small-medium enterprises seeking loans or insurance services. As India advances its E-health agenda, the ABDM strives for an inclusive and consultative approach, gathering diverse perspectives to develop a robust strategy for responsible data sharing.

2.Privacy Preserving Approaches for Healthcare Data Sharing

The advancement of information technology has greatly benefited the medical industry through the implementation of health management systems, which have generated substantial volumes of health big data over time. These health data are exceptionally valuable, as their analysis can uncover significant opportunities for scientific discoveries, innovations, and ultimately, life-saving interventions [13]. However, sharing health data with organizations and institutions for research and analysis purposes raises privacy concerns patient information. Health data contain sensitive and personal details directly tied to individuals, and unauthorized access or misuse can lead to identity theft, denial of health insurance, and exposure to financial fraud.

Approach	**Description**	**Advantages**	**Disadvantages**
Aggregation	De-identification by storing aggregate answers and deleting underlying records.	All underlying data are deleted except true anonymized data.	Requires understanding of initial question in relation to dataset answer. Primarily useful for basic research.
Pseudonymization	Replacing identifiable information with pseudonyms (e.g., replacing "Abiodun Adeoye" with "efg234").	Supports redaction of identifying information and linking data across datasets.	Artificial identifiers need review to prevent identification. Does not anonymize sensitive attributes. Requires careful process control to prevent reverse engineering.

Synthetic data	Generation of data that mimics real data without containing actual individual data.	No privacy risk as it does not include real individual data.	Cannot combine data across datasets about the same individual. Data nuances may be lost, impacting ML and quality improvement tasks. Limited usefulness for applications requiring detailed data characteristics.
Differential privacy	Adding random noise to data until individual identification becomes impractical.	Preserves more information compared to redaction while achieving anonymization.	Complex computation, especially for maintaining utility and desired privacy level. Not for unstructured
			dataset.
Multi-party computing	Collaborative data analysis among multiple parties without sharing datasets directly.	Allows analysis without compromising privacy.	Difficult algorithm development. Slower computation due to cryptographic operations.
			Output values may still reveal identifying information.
Homomorphic encryption	Performing operations on encrypted data without decrypting it, preserving privacy during analysis.	Prevents privacy loss during valuable data analysis.	Complex algorithm development. Slow computation due to cryptographic operations. Output values may reveal identifying information.

Tabel :1 Privacy Preservation Techniques [15]

Privacy-preserving sharing of e-health big data is crucial in the healthcare industry, especially as the sharing of data among healthcare providers and organizations grows rapidly [2]. Health data hold significant value, offering the potential for impactful medical insights and innovations. Conversely, they can additionally be misused for harmful purposes. Ensuring the privacy of patient data is paramount, as these records encompass confidential data that can directly identify individuals. Table 1 provide information of privacy preserving methods and table 2 show comparative information of privacy preserving tools.

Tool	Description	Supported Privacy-Preserving Models	Key Features	Use Cases

ARX	A comprehensive, open-source tool for anonymizing structured personal sensitive data, with cross-platform functionality.	δ-Presence, ℓ-Diversity, k-Anonymity, tCloseness, Average risk, δ-Disclosure privacy, k-Map, βLikeness, Gametheoretic model, Population uniqueness, (ϵ, δ) Differential privacy.	Deletes direct identifiers, limits indirect identifiers Protects sensitive attributes, Handles up to 50 attributes with millions of records, Informative GUI for analysis and visualization	Clinical data sharing Research projects, Commercial big data analytics, Biomedical data analysis, Training purposes
Amnesia	An anonymization tool with a user-friendly graphical interface, allowing hierarchy creation for tailored anonymization.	k-Anonymity, km-Anonymity	Provides data quality statistics Integrates into projects using ReST API Statistical representation of reidentified probabilities	European Union's "My Health My Data" project
OpenDP	An open-source tool for conducting privacypreserving analysis on personal sensitive datasets, using algorithms for differentially private statistical releases.	Differential privacy	Facilitates safe data sharing for scientific research and public interest exploration	Academic institutions Industries Government agencies conducting research and exploration
μ-ARGUS	A tool based on R programming language, designed to create microdata safely, offering statistical analysis with various anonymization techniques.	Randomization, Micro-aggregation, Adding noise, Global recoding, Top and bottom coding, Synthetic data generation	Range of statistical anonymization techniques, Supports synthetic data generation	Statistical analysis

Tabel 2: Privacy Preservation Tools [15]

3.Aim of the Research

The main objective is to investigate and assess different anonymization techniques and algorithms to guarantee the privacy and safety of healthcare data, particularly with the rising adoption of centralized and cloud-based servers.

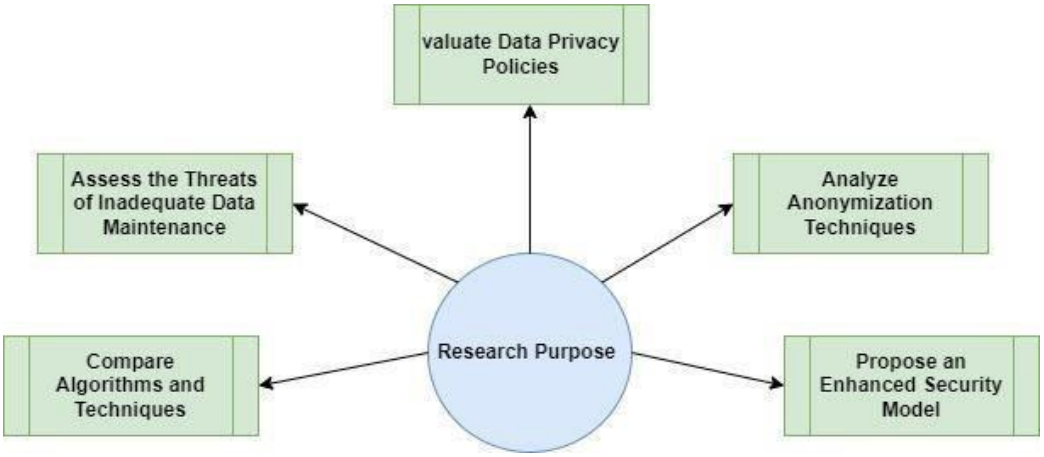

Figure 1: Aim of the Research

4.Proposed Methodology

The algorithm developed here combines three key concepts in privacy of healthcare data and analysis: k-anonymity, n-gram modelling, and differential privacy. It to assure that the released dataset satisfies the k-anonymity criterion, utilizes n-gram modelling for textual analysis, and applies differential privacy for enhanced data protection. Below, we outline the structured algorithm along with the relevant mathematical formulations.

- **Input:** Original dataset D, privacy parameter ϵ for differential privacy, and parameter k for kanonymity.
- **Output:** Anonymized and differentially private dataset D′.

1. Identify quasi-identifiers Q in D that could potentially link data to specific individuals.
2. For each group, generalize quasi-identifiers to guarantee that records within a group are indistinguishable based on Q.
3. D k−anon =Generalize (D, Q, k)
4. For each text entry in column C, generate n-grams.
5. N=n-Gram Frequency (Dk−anon , C, n)
6. Laplace Mechanism: Noise=Lap (Δ f /ϵ)
7. D n−gram′=N+ Noise
8. Replace the original n-gram counts in D k−anon with the noisy counts D’ n−gram.
9. Release the dataset D′D'D′ which meets the requirements of k-anonymity and differential privacy.

This algorithm provides a comprehensive methodology for effectively managing and anonymizing textual data, addressing privacy concerns through a combination of anonymization techniques and statistical noise addition. Each step is designed to protect individuals' privacy represented in the dataset while maintaining its usability for analytical purposes.

5.Experiment Result

In this experiment Adult Dataset [16] used. On dataset we applied prosed privacy anonymization techniques and other traditional privacy techniques. And after that machine learning, model used for predication on anonymized datasets and compared. In Figure 2. Proposed Model excels with the highest accuracy (0.88), precision (0.86), recall (0.89), and F1-score (0.87), suggesting that integrating differential privacy with K-anonymity enhances classification performance. K-anonymity follows with an accuracy of 0.85 and an F1-score of 0.84, showing solid but slightly less balanced performance. Model k-anonymity + AES 256 Bit Encryption, with the lowest accuracy (0.82), precision (0.79), and F1-score (0.82), indicates that AES encryption may introduce noise or data utility loss, although it still maintains a competitive recall (0.85). Overall, proposed proves to be the most reliable and balanced among the three.

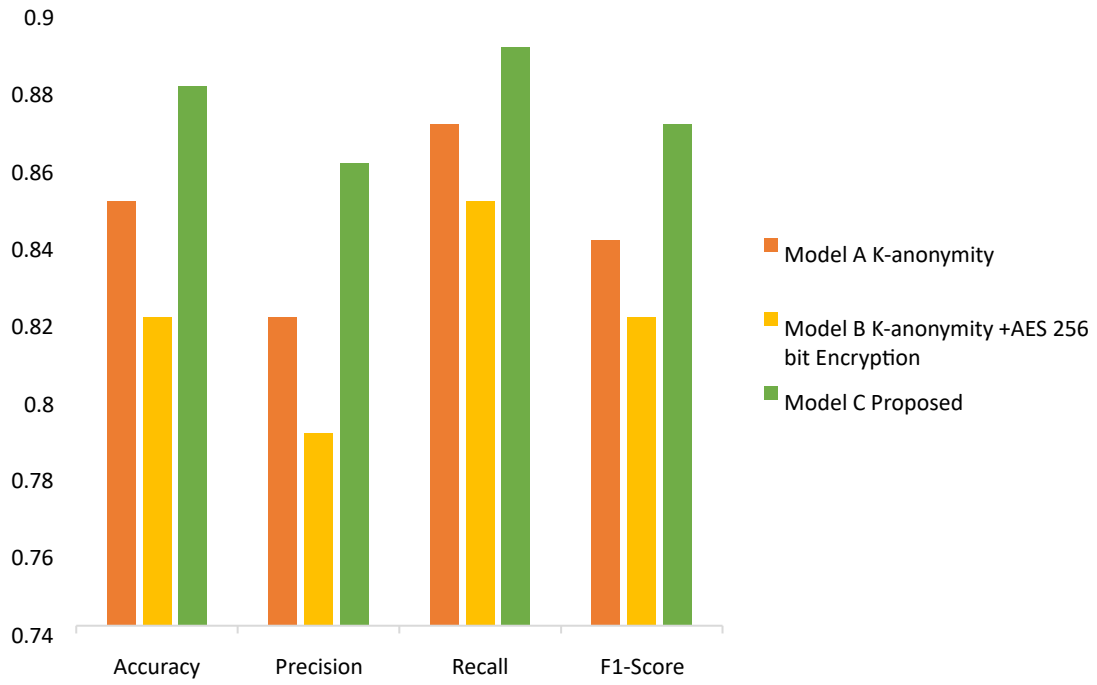

Figure 2. Comparison with Existing Methods

Fold	Accuracy	Precision	Recall	F1-Score
1	0.85	0.82	0.87	0.84
2	0.82	0.79	0.85	0.82
3	0.88	0.86	0.89	0.87
4	0.84	0.81	0.86	0.83
5	0.87	0.84	0.88	0.86

Tabel 3: Cross-Validation Matrix Results

In Tabel 3. each row represents a fold in the cross-validation process, while each column corresponds to a performance metric (accuracy, precision, recall, F1-score). The values in the table reflect the model's performance on each fold, offering insights into its consistency and generalization ability across various subsets of the data. In Figure 3. Data loss during the anonymization process can undermine trust in data protection. If people doubt their data's security, they may be reluctant to share it, hindering research and the implementation of new products and services. The ethical implications of data loss in anonymization are significant, and it's crucial to consider these when deciding how to anonymize data.

In Figure 3. For k-anonymity, utility signifies the extent of information loss, reflecting how much the data has been altered by the anonymization process. For proposed privacy model, utility also represents information loss or accuracy, but it is measured differently because of the probabilistic nature of proposed privacy model, involving the processing of aggregate queries and comparing data distributions before and after applying proposed privacy model. The values highlight the trade-off between privacy & utility. Typically, as privacy increases (by increasing k for k-anonymity or decreasing epsilon for differential privacy), utility tends to decrease.

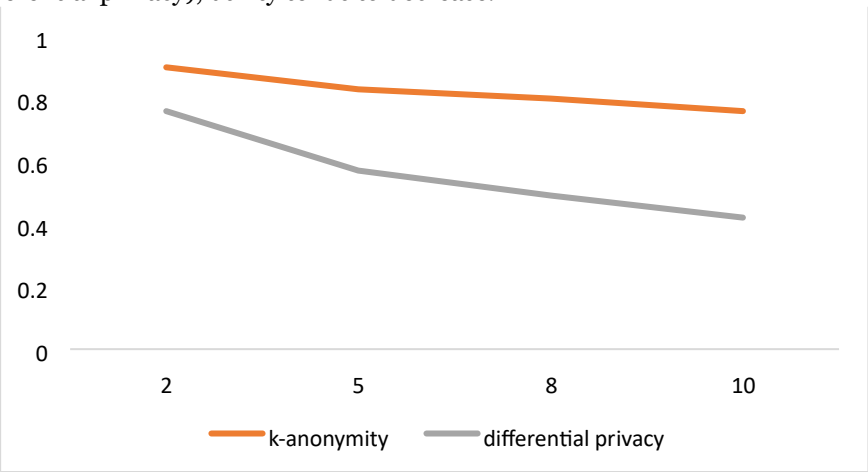

Figure 3. Data utility comparisons (Grey- k-anonymity & Blue-Proposed)

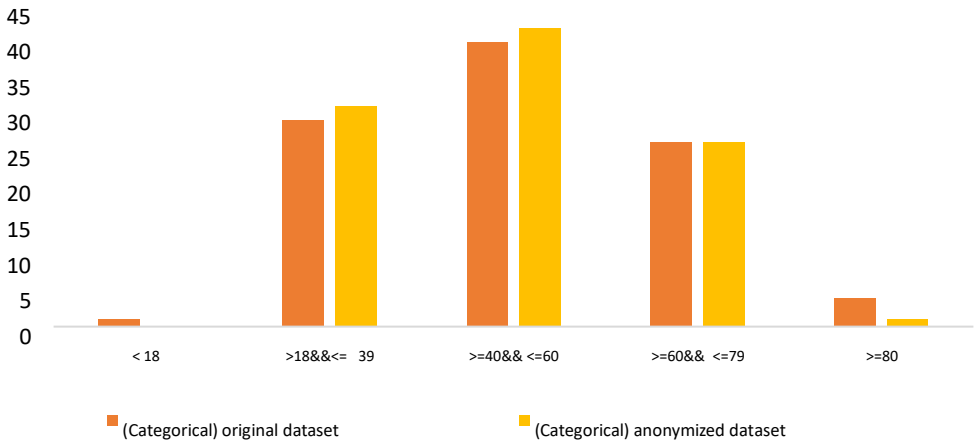

Figure 4. Comparison of Categorical Data After Anonymization

In Figure 4, The anonymization process has preserved the overall distribution patterns of the original dataset. Minor differences are observed in the frequencies of the 18 to 39 and 40 to 60 age groups, indicating slight changes due to anonymization. Frequencies for the under 18 and 80 and above age groups remain consistently low in both datasets, ensuring rare age groups are similarly represented post-anonymization. Overall, the chart demonstrates that the anonymization process has not significantly altered the categorical distribution of age groups, maintained the dataset's integrity and utility while provided privacy protection. Figure 5 represent comparison in terms of time taken to anonymization the data.Figure 7 shows anonymization percentages for different k values (5, 10, and 20) across various dataset sizes (5k; 10k; and 50k) and Qi values (0.5). The percentages reflect the level of anonymity achieved, with higher percentages indicating greater anonymization.

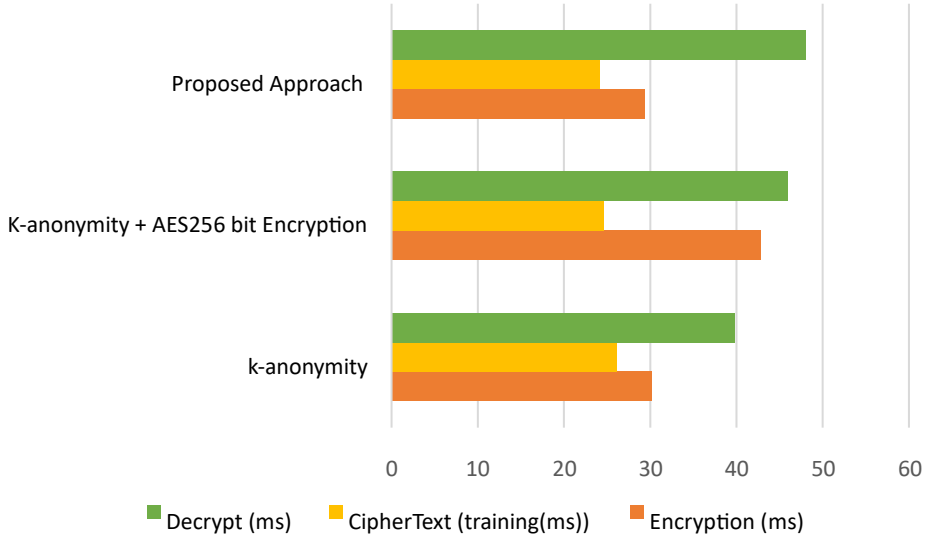

Figure 5. Comparison of Anonymization Timings

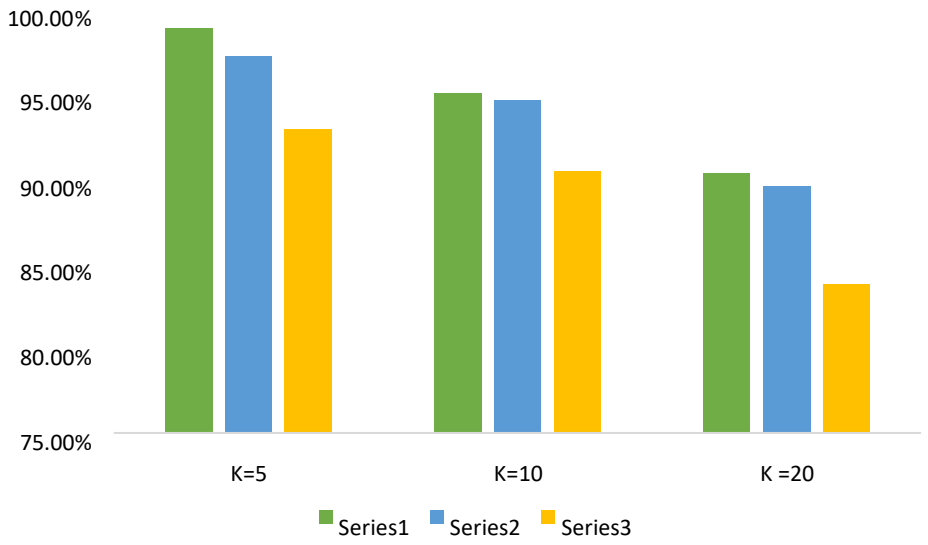

Figure 6. Comparison at Different sensitive parameters

6.Conclusion

The anonymization process has effectively preserved the overall distribution patterns of the original dataset, with only minor variations in the frequencies of certain age groups, thereby maintaining the dataset's integrity and utility while ensuring privacy protection. The utility, reflecting information loss, varies between k-anonymity and the proposed privacy model because to their distinct approaches. As privacy increases, utility generally decreases. This research demonstrates that data loss during anonymization can undermine trust in data protection, potentially impacting data sharing and research. The table of cross-validation results highlights the model's performance consistency and generalization ability. In this paper, with the Adult Dataset, the proposed model, which integrates differential privacy with k-anonymity, outperformed traditional methods, achieving the highest accuracy (0.88), precision (0.86), recall (0.89), and F1-score (0.87). This suggests that the integration enhances classification performance. K-anonymity alone showed solid but slightly less balanced performance, while k-anonymity combined with AES 256-bit encryption had the lowest performance, indicating potential data utility loss due to encryption. Overall, the proposed model demonstrated the highest reliability and balance among the three.

References

1. Seh, A. H., Zarour, M., Alenezi, M., Sarkar, A. K., Agrawal, A., Kumar, R., & Ahmad Khan, R.
2. (2020, May). Healthcare data breaches: insights and implications. In Healthcare (Vol. 8, No. 2, p. 133). MDPI.
3. Murdoch, B. (2021). Privacy and artificial intelligence: challenges for protecting health information in a new era. BMC Medical Ethics, 22, 1-5.
4. Keshta, I., & Odeh, A. (2021). Security and privacy of electronic health records: Concerns and challenges. Egyptian Informatics Journal, 22(2), 177-183.
5. Muthuppalaniappan, M., & Stevenson, K. (2021). Healthcare cyber-attacks and the COVID-19 pandemic: an urgent threat to global health. International Journal for Quality in Health Care, 33(1), mzaa117.
6. Privacy-Preserving in E-health Big Data Sharing | by Abiodun I. Adeoye | Medium(last visit: 12 July 2024)
7. Mehta, D. B. (2023). Privacy-Preserving Machine Learning Architecture for Cross-Channel Advertising Optimization in a Large-Scale Social Media Platform. International Journal of Communication Networks and Information Security (IJCNIS), 15(8), 23–32.
8. Bhosale, K. S., Nenova, M., & Iliev, G. (2021, September). A study of cyber attacks: In the healthcare sector. In 2021 Sixth Junior Conference on Lighting (Lighting) (pp. 1-6). IEEE.
9. Hulsen, T. (2020). Sharing is caring-data sharing initiatives in healthcare. International journal of environmental research and public health, 17(9), 3046.
10. Voigt, P., & Von demBussche, A. (2017). The eu general data protection regulation (gdpr). A Practical Guide, 1st Ed., Cham: Springer International Publishing, 10(3152676), 10-5555.
11. Arpita, M., & Panchal, S. (2022). Smart Health and Cybersecurity in the Era of Artificial Intelligence. In Information and Communication Technology for Competitive Strategies (ICTCS 2021) Intelligent Strategies for ICT (pp. 41-48). Singapore: Springer Nature Singapore.
12. Sharma, R. S., Rohatgi, A., Jain, S., & Singh, D. (2023). The Ayushman Bharat Digital Mission (ABDM): making of India's digital health story. CSI Transactions on ICT, 11(1), 3-9.
13. Aayog, N. I. T. I. (2020). Data empowerment and protection architecture. Draft for Discussion, 36-37.
14. Maheriya, A., & Panchal, S. (2023). A Neoteric Approach to Improvise Privacy Shielding of Sensitive Healthcare Information & Prediction of Disease. International Journal of Computing and Digital Systems, 13(1), 1-12.
15. Raval, A., & Maheriya, A. (2023). A Survey on Deep Learning Methods for Addressing COVID19 Issues. In Information and Communication Technology for Competitive Strategies (ICTCS 2022) Intelligent Strategies for ICT (pp. 61-73). Singapore: Springer Nature Singapore.
16. Adult income dataset (kaggle.com)(last visit: 12 July 2024)
17. Maheriya, A. A Heuristic Approach to Protect Privacy of Patient's Sensitive Data with Prediction of Disease.
18. Act, A. (1996). Health insurance portability and accountability act of 1996. Public law, 104, 191.
19. Act, P. (2000). Personal information protection and electronic documents act. Department of
20. Justice, Canada. Full text available at http://laws. justice. gc. ca/en/P-8.6/text. html, 43564364.
21. Arpita, M., & Panchal, S. (2022). Smart Health and Cybersecurity in the Era of Artificial Intelligence. In Information and Communication Technology for Competitive Strategies (ICTCS 2021) Intelligent Strategies for ICT (pp. 41-48). Singapore: Springer Nature Singapore.
22. Sharma, R. S., Rohatgi, A., Jain, S., & Singh, D. (2023). The Ayushman Bharat Digital Mission (ABDM): making of India's digital health story. CSI Transactions on ICT, 11(1), 3-9.
23. Aayog, N. I. T. I. (2020). Data empowerment and protection architecture. Draft for Discussion, 36-37.
24. Seh, A. H., Zarour, M., Alenezi, M., Sarkar, A. K., Agrawal, A., Kumar, R., & Ahmad Khan, R.
25. (2020, May). Healthcare data breaches: insights and implications. In Healthcare (Vol. 8, No. 2, p. 133). MDPI.
26. Murdoch, B. (2021). Privacy and artificial intelligence: challenges for protecting health information in a new era. BMC Medical Ethics, 22, 1-5.
27. Keshta, I., & Odeh, A. (2021). Security and privacy of electronic health records: Concerns and challenges. Egyptian Informatics Journal, 22(2), 177-183.
28. Muthuppalaniappan, M., & Stevenson, K. (2021). Healthcare cyber-attacks and the COVID-19 pandemic: an urgent threat to global health. International Journal for Quality in Health Care, 33(1), mzaa117.
29. Maheriya, A., & Panchal, S. (2023). A Neoteric Approach to Improvise Privacy Shielding of Sensitive Healthcare Information & Prediction of Disease. International Journal of Computing and Digital Systems, 13(1), 1-12.

30. Raval, A., & Maheriya, A. (2023). A Survey on Deep Learning Methods for Addressing COVID19 Issues. In Information and Communication Technology for Competitive Strategies (ICTCS 2022) Intelligent Strategies for ICT (pp. 61-73). Singapore: Springer Nature Singapore.

31. Privacy-Preserving in E-health Big Data Sharing | by Abiodun I. Adeoye | Medium(last visit: 12 July 2024)

32. Adult income dataset (kaggle.com)(last visit: 12 July 2024)

33. Maheriya, A. A Heuristic Approach to Protect Privacy of Patient's Sensitive Data with Prediction of Disease.

34. Bhosale, K. S., Nenova, M., & Iliev, G. (2021, September). A study of cyber attacks: In the healthcare sector. In *2021 Sixth Junior Conference on Lighting (Lighting)* (pp. 1-6). IEEE.

35. Hulsen, T. (2020). Sharing is caring—data sharing initiatives in healthcare. *International journal of environmental research and public health*, *17*(9), 3046.

36. Voigt, P., & Von demBussche, A. (2017). The eu general data protection regulation (gdpr). *A Practical Guide, 1st Ed., Cham: Springer International Publishing*, *10*(3152676), 10-5555.

37. Act, A. (1996). Health insurance portability and accountability act of 1996. *Public law*, *104*, 191.

38. Act, P. (2000). Personal information protection and electronic documents act. *Department of Justice, Canada. Full text available at http://laws. justice. gc. ca/en/P-8.6/text. html*, 43564364.